Jingjing Wu

CURRICULUM VITAE

POSTDOCTORAL ASSOCIATE · DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

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Education

Clarkson University, Department of Civil and Environmental Engineering — *Potsdam*

PHD IN CIVIL AND ENVIRONMENTAL ENGINEERING — *2015 - 2021*

Advisor: Dr. Shane W. Rogers

Dissertation Topic: "Bioextractive Aquaculture as an Alternative Nutrient Management Strategy for Water Resource Recovery Facilities: Comprehensive Evaluation Based on Techno-Economic Analysis and Life Cycle Assessment"

Clarkson University, Department of Civil and Environmental Engineering — *Potsdam*

MASTER OF SCIENCE IN CIVIL AND ENVIRONMENTAL ENGINEERING — *2013 - 2015*

Advisor: Dr. Shane W. Rogers

Ocean University of China, College of Marine Life Sciences — *Qingdao*

BACHELOR OF SCIENCE IN BIOLOGICAL SCIENCE — *2008 - 2012*

Professional Credentials

2026 **Engineer-in-Training (EIT)**, Texas Board of Professional Engineers and Land Surveyors

2026 **PE Environmental Exam - Passed**, NCEES Principles and Practice of Engineering (PE) exam

Publications

Peer-Reviewed Publications

Kelley Plymesser, **Jingjing Wu**, Rachel M. West, Lauren B. Stadler. Measles Wild-Type Virus Detection Through Wastewater Surveillance in Sandoval County, New Mexico. *JAMA Network Open (2026)*

Jingjing Wu, Michael X. Wang, Todd J. Treangen, Katherine B. Ensor, Loren Hopkins, Lauren B. Stadler. Sensitivity, throughput, and cost analysis of concentration methods for multi-target pathogen wastewater monitoring. *ACS ES&T Water (2025)*

Jingjing Wu, Katherine B. Ensor, Loren Hopkins, Lauren B. Stadler. Assessment and Application of GeneXpert Rapid Testing for Respiratory Viruses in School Wastewater. *Environmental Science: Water Research & Technology (2025)*

Sarah Javornik Cregeen, Mike Tisza, Blake Hanson, Marissa Cook, Anil Surathu, Rebecca Schneider, **Jingjing Wu**, Kirstin Short, Kaavya Domakonda, Loren Hopkins, Matt Ross, Joseph Petrosino, Jennifer Deegan, Lauren B. Stadler, Eric Boerwinkle, Anthony Maresso. Sequencing-based Detection of Measles in Wastewater. *American Journal of Public Health (2025)*

Jingjing Wu, Michael X. Wang, Prashant Kalvapalle, Michael Nute, Todd J. Treangen, Katherine Ensor, Loren Hopkins, Rachel Poretsky, Lauren B. Stadler. Multiplexed detection, partitioning, and persistence of wild type and vaccine strains of measles, mumps, and rubella viruses in wastewater. *Environmental Science & Technology (2024)*

Katherine B. Ensor, Julia C. Schedler, Thomas Sun, Rebecca Schneider, Anthony Mulenga, **Jingjing Wu**, Lauren B. Stadler, Loren Hopkins. Online trend estimation and detection of trend deviations in sub-sewershed time series of SARS-CoV-2 RNA measured in wastewater. *Scientific Reports (2024)*

Jingjing Wu, Shane W. Rogers, Rebekah Schaummann, Nichole Price. A Comparison of multiple macroalgae cultivation systems and end-use strategies of Saccharina latissima and Gracilaria tikvahiae based on techno-economic analysis and life cycle assessment. *Sustainability (2023)*

Jingjing Wu, Shane W. Rogers, Rebekah Schaummann, Chris Higgins, Nichole Price. Bioextractive aquaculture as an alternative nutrient management strategy for water resource recovery facilities. *Water Research (2022)*

Preprints

Jingjing Wu, Michael X. Wang, Kaavya Domakonda, Rebecca Schneider, Kirstin Short, Charlene Offiong, Todd J. Treangen, Katherine Ensor, Loren Hopkins, Lauren B. Stadler. Tracking *Candida auris* in Communities via Wastewater: Facility-Level Surveillance and Targeted Sequencing. *medRxiv (2026)*

Professional Experience

Postdoctoral Associate *2022-Present*

Department of Civil and Environmental Engineering, Rice University

Houston, TX, U.S.A. (Advisor: Dr. Lauren B. Stadler)
Research summary: Develop and validate ddPCR and targeted amplicon sequencing assays for wastewater pathogen surveillance. Compare sampling strategies, sample-processing methods, and molecular detection workflows to optimize multi-target wastewater surveillance pipelines and guide method selection. Current research focuses on using wastewater metagenomic data to inform targeted amplicon assay design and combining laboratory experiments with machine learning to characterize and predict the effects of sequence mismatches on PCR performance.

Postdoctoral Research Associate *2021-2022*

Department of Civil and Environmental Engineering, Clarkson University

Potsdam, NY, U.S.A. (Advisor: Dr. Shane W. Rogers)
Research summary: Developed preliminary techno-economic and life cycle assessment models to evaluate the potential benefits of bipolar membrane electrodialysis for nutrient recovery from anaerobic digestion sidestreams.

Visiting Scientist *Mar.-May 2022*

Bigelow Laboratory for Ocean Sciences

East Boothbay, Maine, U.S.A.
Research summary: Conducted laboratory experiments to assess how nitrogen and phosphorus concentrations in seawater affect seaweed growth.

Graduate Research Assistant *2013-2021*

Department of Civil and Environmental Engineering, Clarkson University

Potsdam, NY, U.S.A.
Research summary: Integrated wastewater process modeling with life cycle assessment and techno-economic analysis to evaluate nutrient removal and macroalgae-based bioextraction strategies. Conducted bench-scale experiments to evaluate anaerobic co-digestion and the removal of fecal indicators by reverse osmosis.

Teaching & Mentoring Experience

Teaching

Spring 2025, 2026	**Teaching Assistant**, CEVE 316 Urban Water Systems Lab – Process Modeling Module, Rice University
Spring 2025	**Workshop Instructor**, Wastewater Surveillance for Pathogens: Data Generation and Application, Rice University Environmental Data Science Academy, Rice University *Highlight: Designed an interactive Google Colab module to visualize the relationship between community transmission and wastewater surveillance signals.*
Fall 2020	**Guest Lecturer**, CE 479/579 Water and Wastewater Treatment Design, Clarkson University *Highlight: Guided students in developing a simplified model of the local wastewater treatment plant using GPS-X Pro.*
Spring 2019, 2020	**Teaching Assistant**, ES330 Fluid Mechanics, Clarkson University
Fall 2019	**Teaching Assistant**, CE 479/579 Water and Wastewater Treatment Design, Clarkson University
Fall 2018	**Guest Lecturer**, CE 479/579 Water and Wastewater Treatment Design, Clarkson University

Mentoring and Research Advising

2026 **Julia Lim**, Undergraduate researcher, Rice University
2025 **Angelika Contreras**, NSF-RET mentee, HUMBLE ISD
2024 **Tobenna Esomeju**, Undergraduate researcher, Rice University
2024 **Christian Diaz**, Undergraduate researcher, Rice University
2019-2022 **Rebekah Schaummann**, Honors program student, Clarkson University
2018-2020 **Madison Bredehoeft**, Honors program student, Clarkson University
2019 **Angelika Sonne**, NSF-REU student, California State University, San Diego
2018 **Sasha Gallimore-Repole**, Honors program student, Clarkson University
2018 **Alexandra Hauke**, NSF-REU student, North Carolina A&T State University
2017 **Paige Hval**, NSF-REU student, Loyola Marymount University

Presentations

Oral Presentations

2025 *Tracking Candida auris in communities via wastewater: Facility-level surveillance and targeted sequencing*, Southeast Regional Wastewater Surveillance Workshop, Houston, TX, U.S.A.

2024 *Wastewater surveillance for measles: Assay development, application, and action plan*, NSF RCN webinar, Virtual *(Invited)*

2023 *Houston's wastewater monitoring of SARS-CoV-2, influenza, RSV, and mpox for infectious disease surveillance*, SIMB Annual Meeting, Minneapolis, MN, U.S.A. *(Invited)*

2022 *Comparison of multiple macroalgae cultivation systems and end-use strategies based upon life cycle assessment*, Coast-Cow-Consumer 1st Friday Research Forum, Virtual *(Invited)*

2022 *Comparison of multiple macroalgae cultivation systems and end-use strategies based upon technoeconomic analysis and life cycle assessment*, ASLO Aquatic Sciences Meeting, Virtual

2021 *Nutrients and pathogen export from application of manure to frozen ground*, DEMOCRATIA-AQUA-TECHNICA Conference, Virtual

2019 *Seaweed cultivation as an alternative nutrient management strategy for wastewater treatment and recovery facilities: a case study in Boothbay Harbor, Maine*, ASLO Aquatic Sciences Meeting, San Juan, PR, U.S.A.

2018 *Impact of multiple wastes on biomethane potential of anaerobic co-digestion of cow manure*, New York Water Environment Association Conference Annual Meeting & Exhibition, New York City, NY, U.S.A.

Posters

2025 *Wastewater surveillance for Candida auris in nursing homes and downstream wastewater treatment plants in Houston, TX*, Association of Environmental Engineering & Science Professors (AEESP) Research and Education Conference, Durham, NC, U.S.A.

2024 *Multiplexed detection, partitioning, and persistence of wild type and vaccine strains of measles, mumps, and rubella viruses in wastewater*, Wastewater Disease Surveillance Summit, Atlanta, GA, U.S.A.

2022 *Bioextractive macroalgae cultivation for nutrient management as a potential alternative to advanced wastewater treatment nutrient removal processes in water resource recovery facilities*, ASLO Aquatic Sciences Meeting, Virtual

2018 *Macroalgae cultivation in wastewater discharge plumes for nutrient management as a potential alternative to wastewater treatment nutrient removal processes*, North East Graduate Student Water Symposium (NEGSWS), Amherst, MA, U.S.A.

Honors & Awards

2021 **The Phalanx Distinguished Service Award**, Clarkson University

2010 **Student Research Training Program Fellowship**, Ocean University of China

Professional Service & Community Engagement

Professional Service

2026 **Facilitator**, wet-lab breakout session, Pathogen Genomics & Wastewater Surveillance Technical Working Meeting, held at the CDC Roybal Campus, Atlanta, GA.

2026 **Co-host**, "R Study Hall" for the National Wastewater Surveillance System (NWSS) community, virtual.

2025-2026 **Grant reviewer**, Renewable Natural Resources Foundation. Reviewed 10 letters of inquiry in 2026 and 6 proposals in 2025.

2022; 2024–2026 **Journal reviewer**, Water Research, Environmental Science & Technology, International Journal of Infectious Diseases, Green Chemistry, RSC Sustainability, ACS ES&T Water, Environmental Science: Water Research & Technology, Discover Applied Sciences.

2024 **Student presentation evaluator**, Rice University SSPB Summer Research Symposium

2022 **Student presentation evaluator**, ASLO Aquatic Sciences Meeting

Community Engagement

2020-2021 **Graduate Researcher**, COVID-19 surveillance in diverse communities in upstate New York, conducted at Clarkson University in Potsdam, NY

2018 **Assistant Workshop Organizer**, Small-scale Anaerobic Digester Workshop: Farmer education and outreach program regarding the potential use of anaerobic digestion on small farms, conducted at the Cornell Cooperative Extension Farm in Canton, NY